

# Structure-aware Fourier modal analysis: per-harmonic reductions and exact eigenproblem decompositions for fast rigorous coupled-wave analysis

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## Abstract

Rigorous coupled-wave analysis (RCWA) [MG81; Moh+95], also known as the Fourier modal method, expands the fields of a layered periodic structure in a truncated set of  $n$  Fourier harmonics and assembles a scattering ( $S$ ) matrix from a  $2n \times 2n$  non-Hermitian eigenproblem and from dense Redheffer (star) products over the layer stack [Red62; Li96a]. The generic implementation therefore costs  $O(n^3)$  per layer interface and  $O(n^3)$  for the eigensolve, which becomes prohibitive for the many-harmonic problems of modern applications (metasurfaces, EUV lithography masks, multilayer mirrors), where  $n$  can reach hundreds to thousands.

We show that most of this cubic cost is avoidable by exploiting algebraic structure that is already present in the governing equations. (i) *Uniform layers*: a homogeneous layer never couples different diffraction orders; under a per-harmonic  $(E_{x,h}, E_{y,h})$  interleaving, its operator is block-diagonal with  $2 \times 2$  blocks for *arbitrary* truncation sets and *arbitrary* oblique incidence. The  $S$ -matrix of any all-uniform multilayer stack (incident medium, caps, multilayer mirrors, substrates) is therefore built by independent per-harmonic  $2 \times 2$  recursions in  $O(n N_{\text{layers}})$ , eliminating the cubic cost of the stack. (ii) *Patterned layer*: the  $2n \times 2n$  eigenproblem decouples exactly into two  $n \times n$  problems — block-diagonal at normal incidence for mirror-symmetric patterns, and block-lower-triangular at oblique incidence for  $y$ -invariant patterns, the latter with an exact rank-structured reconstruction of the coupled eigenvectors. (iii) *Binary gratings*: the permittivity convolution matrix is diagonal-plus-low-rank (a discrete prolate spheroidal concentration), which bounds the active eigen-subspace and enables reduced-rank eigensolves. (iv) *Post-processing*: the volume integral is evaluated block-wise without materializing  $4n \times 4n$  matrices.

The resulting solver reproduces the reference implementations grcwa and S4 [LF14] to machine precision and reduces the wall-clock time of a full-2D EUV absorber stack on a  $40 \times (\text{Mo/Si})$  mirror by  $3.5 \times$  (the multilayer suffix alone dropping from hours-scale dense products to microseconds), and by up to  $30 \times$  end-to-end for quasi-1D geometries.

## 1 Introduction

### 1.1 Motivation

Rigorous coupled-wave analysis (RCWA), also called the Fourier modal method (FMM), is the workhorse for computing the optical response of layered, laterally periodic photonic structures: diffraction gratings, photonic crystals, thin-film solar cells, metasurfaces, and the phase-shifting absorber stacks of extreme-ultraviolet (EUV) lithography [MG81; Moh+95; LF14]. The method expands the electromagnetic field in a truncated set of  $n$  planewave (Fourier) harmonics in the plane of periodicity, reduces Maxwell's equations in each laterally structured layer to a  $2n \times 2n$  non-Hermitian eigenproblem, and propagates the modal amplitudes across the stack with a scattering-matrix (Redheffer) recursion [WC99; Li96a]. Reflectivity, transmissivity, per-order

diffraction efficiencies, near-field intensities, and absorbed-power distributions all follow from the assembled  $S$ -matrix and the per-layer modal data.

Two application trends drive the need for speed. First, the number of retained harmonics  $n$  must grow to resolve fine features: an EUV mask absorber (60 nm feature in a 300 nm cell) requires  $n \sim 10^2$ ; a line grating of period  $3.5 \mu\text{m}$  supports  $\pm 259$  propagating orders alone, and its quasi-2D simulation (with a handful of coupled  $y$ -orders) needs  $n \sim 10^3$ – $10^4$ . Second, optimization and sensitivity studies — mask co-optimization, tolerance analysis, inverse design [1] — evaluate the forward solver at hundreds to thousands of parameter points, so a  $10$ – $100\times$  speedup per solve is directly multiplied into the total study time.

## 1.2 The cubic cost and why it hurts

A generic implementation spends its time in two places, both cubic in  $n$ :

1.  $O((2n)^3)$  for the (generally non-Hermitian, complex) eigendecomposition (8) of each laterally structured layer, and
2.  $O((2n)^3)$  dense matrix products and inversions *per layer interface* in the Redheffer recursion (12).

To fix scales, consider a routine EUV-mask stack: vacuum | TaN absorber | Ru cap |  $40\times(\text{Mo/Si})$  mirror | Si substrate, i.e. one patterned layer and  $\approx 80$  homogeneous interfaces. With a quasi-2D truncation of  $n \approx 3500$  harmonics ( $N = 2n \approx 7000$ ), one dense complex  $N \times N$  product costs  $\approx N^3 \approx 3.4 \times 10^{11}$  complex operations. Each interface contributes on the order of ten such products plus an  $LU$  solve, and the 80 interfaces of the mirror dominate the budget at  $\gtrsim 10^{14}$ – $10^{15}$  operations — on the order of hours even on a multicore workstation. The patterned-layer eigenproblem contributes a further  $\sim 25N^3$  operations. Any reduction that removes the *number* of interfaces, or the *dimension* against which the stack is processed, is thus worth far more than a constant-factor tuning of a single dense product.

## 1.3 Related work

The Fourier-modal method was introduced for diffraction gratings by Moharam and Gaylord [MG81] and refined into a stable, efficient formulation for binary gratings in [Moh+95]. The stability of the layer-recursion step was settled by Li’s comparison of the  $R$ -,  $T$ -, and  $S$ -matrix algorithms [Li96a], establishing the scattering-matrix (Redheffer) formulation as the method of choice for thick and lossy stacks; our work uses exactly this recursion [Red62]. The convergence of the modal expansion for discontinuous media was analyzed by Li [Li96b], whose inverse rule (and its generalization, the normal-vector field / fast Fourier factorization [Li97]) removes the algebraic convergence bottleneck of the naive Laurent rule for high-contrast metallic gratings. The scattering-matrix treatment of patterned multilayer photonic structures was systematized by Whittaker and Culshaw [WC99].

The performance of FMM solvers has been improved along several axes: better truncation and basis selection, factorization rules that reduce the required  $n$  at fixed accuracy [Li96b; Li97], and — in the present context — the exploitation of *structure within the algebraic operators*. Two structural facts are classical for one-dimensional gratings: the  $S$ -matrix of a stack of homogeneous layers decouples into independent scalar (or  $2 \times 2$ ) per-order problems [WC99], and at normal incidence the tangential TE/TM polarizations decouple. The present paper makes three contributions beyond these classical facts: it (i) proves and exploits the per-harmonic decoupling for *arbitrary* two-dimensional harmonic sets and *oblique* incidence (Section 3); (ii) derives *exact* decompositions of the patterned-layer eigenproblem — block-diagonal at normal incidence, and block-lower-triangular at oblique incidence for  $y$ -invariant gratings, with an explicit reconstruction of the coupled eigenvectors (Section 4); and (iii) identifies a low-rank (diagonal-plus-low-rank) structure in the permittivity convolution matrix of binary gratings, inherited from the eigenvalue concentration of the discrete prolate spheroidal matrix [Sle78] (Section 6). We also give a block-wise evaluation of the volume integral (Section 7) and a validated, heavily optimized implementation (Sections 8–9).

Finally, we note the role of reference implementations. The open-source RCWA codes `grcwa` [ ] (a differentiable Python implementation) and `S4` [LF14] serve as our golden references: every optimization in this paper is verified to reproduce their scattering observables to machine precision, so the reductions provably do not alter the physics.

## 1.4 Contributions

Summarized, the contributions of this paper are:

1. *Per-harmonic uniform-suffix factorization*. We prove (Lemma 1) that a homogeneous layer’s operator is  $2 \times 2$ -block-diagonal per harmonic under an interleaving of the field components, for arbitrary truncation sets and arbitrary oblique incidence, and (Corollary 1) that the  $S$ -matrix of any all-homogeneous stack is therefore  $2 \times 2$ -block-diagonal. The suffix is assembled by independent per-harmonic  $2 \times 2$  Redheffer recursions in  $O(n N_{\text{layers}})$ , reducing the multilayer mirror from hours-scale to microseconds.
2. *Exact eigenproblem decompositions*. For the patterned layer we prove the block-diagonal (normal incidence, mirror-symmetric patterns) and block-lower-triangular (oblique incidence,  $y$ -invariant gratings) forms of  $\mathbf{M}$ , each splitting the  $2n \times 2n$  problem into two  $n \times n$  ones at  $4\times$  fewer eigensolver flops, and give the exact reconstruction (31) of the coupled eigenvectors.
3. *TE/TM-decoupled scattering path* (Section 5): a further  $4\times$  reduction on the patterned prefix and assembly at normal incidence.
4. *Low-rank structure of binary gratings* (Section 6): closed-form Fourier coefficients and the diagonal-plus-low-rank form of both patterned eigenproblems, enabling reduced-rank eigensolves.
5. *Block-wise post-processing* (Section 7): the absorption and stress-tensor integrals without  $4n \times 4n$  /  $3n \times 3n$  matrices.
6. *A validated high-performance implementation* (Sections 8–9), bit-identical to the reference solvers and  $3.5\times$ – $30\times$  faster on representative stacks.

All reductions are *exact*: they change only the order of arithmetic, not the discretized model. This is in contrast to convergence-accelerating techniques [Li96b; Li97], which reduce the required  $n$ ; the two families are orthogonal and compound.

## 1.5 Organization

The remainder of the paper is organized as follows. Section 2 derives the Fourier-modal formulation from Maxwell’s equations and fixes notation. Section 3 proves the per-harmonic structure of homogeneous layers and develops the suffix algorithm. Section 4 derives the exact eigenproblem decompositions of the patterned layer, with complete proofs. Section 5 treats the TE/TM-decoupled path, Section 6 the low-rank structure of binary gratings, and Section 7 the block-wise post-processing. Section 8 collects algorithmic implementation details, Section 9 validates and benchmarks, and Section 10 discusses limits and outlook.

# 2 Fourier-modal formulation

## 2.1 Geometry, harmonics, and notation

Consider a stack of layers, each invariant along  $z$  (the propagation axis), with the in-plane permittivity  $\varepsilon(x, y)$  periodic on a lattice with direct vectors  $\mathbf{L}_1, \mathbf{L}_2$  and reciprocal vectors  $\mathbf{K}_1, \mathbf{K}_2$ . The field is expanded in the  $n$

retained harmonics  $\{\mathbf{G}_i\}_{i=1}^n \subset \{m_1\mathbf{K}_1 + m_2\mathbf{K}_2\}$ , so that every in-plane quantity is represented by  $n$  Fourier coefficients and the transverse wavevectors of harmonic  $i$  are

$$k_{x,i} = k_{x0} + (m_1\mathbf{K}_1 + m_2\mathbf{K}_2) \cdot \hat{x}, \quad k_{y,i} = k_{y0} + (m_1\mathbf{K}_1 + m_2\mathbf{K}_2) \cdot \hat{y}, \quad (1)$$

with  $k_{x0}, k_{y0}$  fixed by the incidence angles. We write  $\mathbf{k}_x, \mathbf{k}_y \in \mathbb{C}^n$  for the diagonal entries  $k_{x,i}, k_{y,i}$ , and let  $\mathbf{q} \in \mathbb{C}^{2n}$  denote the (unknown) vertical propagation constants. The total block size is  $N = 2n$ ; all scattering matrices are  $N \times N$ .

## 2.2 The layer eigenproblem

We derive the eigenproblem from Maxwell's equations. With the  $e^{+i\omega t}$  time convention, the curl equations in a source-free, non-magnetic medium are

$$\nabla \times \mathbf{E} = -i\omega\mathbf{H}, \quad \nabla \times \mathbf{H} = i\omega\varepsilon\mathbf{E}. \quad (2)$$

Fourier-expanding the transverse field components  $e_x, e_y, h_x, h_y \in \mathbb{C}^n$  and stacking  $\mathbf{e}_t = [e_x; e_y]$ ,  $\mathbf{h}_t = [h_x; h_y]$  (in the ordering  $[E_{x,1} \dots E_{x,n}; E_{y,1} \dots E_{y,n}]$ ), the in-plane derivatives  $\partial_x, \partial_y$  become multiplications by the diagonal matrices  $K_x = \text{diag}(k_{x,i})$ ,  $K_y = \text{diag}(k_{y,i})$ , and the permittivity becomes the convolution operator  $\mathcal{C}$ . Eliminating the longitudinal components from (2),

$$e_z = -\frac{1}{\omega} \mathcal{C}^{-1}(K_y h_x - K_x h_y), \quad h_z = \frac{1}{\omega} (K_y e_x - K_x e_y), \quad (3)$$

and rearranging the transverse components yields the first-order system

$$\frac{d}{dz} \begin{bmatrix} e_x \\ e_y \\ h_x \\ h_y \end{bmatrix} = \mathbf{A} \begin{bmatrix} e_x \\ e_y \\ h_x \\ h_y \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{0} \end{bmatrix}, \quad (4)$$

with the  $2n \times 2n$  blocks

$$\mathbf{A}_{12} = \begin{bmatrix} -\frac{i}{\omega} K_x \mathcal{C}^{-1} K_y & -i\omega I + \frac{i}{\omega} K_x \mathcal{C}^{-1} K_x \\ i\omega I - \frac{i}{\omega} K_y \mathcal{C}^{-1} K_y & \frac{i}{\omega} K_y \mathcal{C}^{-1} K_x \end{bmatrix}, \quad (5)$$

$$\mathbf{A}_{21} = \begin{bmatrix} \frac{i}{\omega} K_x K_y & i\omega \mathcal{C} - \frac{i}{\omega} K_x^2 \\ -i\omega \mathcal{C} + \frac{i}{\omega} K_y^2 & -\frac{i}{\omega} K_y K_x \end{bmatrix}. \quad (6)$$

Because  $\mathbf{A}_{12}\mathbf{A}_{21}$  and  $\mathbf{A}_{21}\mathbf{A}_{12}$  share their eigenvalues, the eigenvalues of  $\mathbf{A}$  occur in  $\pm$ -pairs,  $\{\pm iq_j\}$ , and eliminating the magnetic components gives the second-order equation

$$\frac{d^2 \mathbf{e}_t}{dz^2} = -\mathbf{A}_{12}\mathbf{A}_{21} \mathbf{e}_t, \quad (7)$$

whose eigenvalues are  $-q_j^2$ . In the planewave (Fourier-modal) basis the same problem is written as the  $2n \times 2n$  non-Hermitian eigenproblem

$$\mathbf{M} \Phi = \Phi \text{diag}(\mathbf{q}^2), \quad (8)$$

whose eigenvalues are the squared vertical propagation constants  $q_j^2$ , and the modal fields vary as  $e^{\pm iq_j z}$ . The matrix  $\mathbf{M}$  is built from the permittivity convolution (Toeplitz) matrix  $\mathcal{C}_{ij} = \hat{\varepsilon}(\mathbf{G}_i - \mathbf{G}_j)$ ,

$$\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top, \quad \text{ep2} = \text{blkdiag}(\mathcal{C}, \mathcal{C}), \quad (9)$$

with  $\mathbf{E} = \mathcal{C}^{-1}$ , and

$$\mathbf{K} = \omega^2 \mathbf{I} - (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top, \quad \mathbf{J}_k = \begin{bmatrix} -\text{diag}(\mathbf{k}_y) \\ \text{diag}(\mathbf{k}_x) \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} \text{diag}(\mathbf{k}_x) \\ \text{diag}(\mathbf{k}_y) \end{bmatrix}. \quad (10)$$

The branch cut  $\mathbf{q}_i \mapsto -\mathbf{q}_i$  when  $\text{Im } \mathbf{q}_i < 0$  selects the physically decaying (evanescent) modes in  $+z$ . Here  $\omega$  is the (possibly complex) angular frequency; the lossy case  $\text{Im } \omega \neq 0$  or complex  $\varepsilon$  is included throughout. For a homogeneous layer of permittivity  $\varepsilon_0$ ,  $\mathbf{E} = 1/\varepsilon_0$  and the same formulas apply with  $\mathcal{C}$  replaced by the scalar  $\varepsilon_0$ . (The placement of  $\mathbf{E}$  in (10) implements the inverse-rule / Lippmann–Schwinger factorization of the discontinuous TM component [Li96b], which is essential for convergence with metallic gratings.)

### 2.3 S-matrix (Redheffer) recursion

Two adjacent layers  $l, l+1$  are connected by the interface matrices

$$\mathbf{Q}_l = \Phi_l^{-1} \Phi_{l+1}, \quad \mathbf{P}_l = \text{diag}(\mathbf{q}_l) (\mathbf{K}_l \Phi_l)^{-1} \mathbf{K}_{l+1} \Phi_{l+1} \text{diag}(1/\mathbf{q}_{l+1}), \quad (11)$$

and the transmission matrices  $\mathbf{T}_{11} = \frac{1}{2}(\mathbf{Q} + \mathbf{P})$ ,  $\mathbf{T}_{12} = \frac{1}{2}(\mathbf{Q} - \mathbf{P})$ . The scattering matrix of the composite stack follows from the Redheffer star product [Red62; Li96a]: with diagonal phase factors  $\mathbf{d}_\ell = \text{diag}(e^{iq_\ell t_\ell})$  for layer thickness  $t_\ell$ , one update of the accumulated  $S$ -matrix is

$$\begin{aligned} \mathbf{P}_1 &= (\mathbf{T}_{11} - \mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{12})^{-1}, \\ \mathbf{S}_{11} &\leftarrow \mathbf{P}_1 \mathbf{d}_1 \mathbf{S}_{11}, \quad \mathbf{S}_{12} \leftarrow \mathbf{P}_1 (\mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{11} - \mathbf{T}_{12}) \mathbf{d}_2, \\ \mathbf{S}_{21} &\leftarrow \mathbf{S}_{21} + \mathbf{S}_{22} \mathbf{T}_{12} \mathbf{S}_{11}, \quad \mathbf{S}_{22} \leftarrow \mathbf{S}_{22} (\mathbf{T}_{11} \mathbf{d}_2 + \mathbf{T}_{12} \mathbf{S}_{12}). \end{aligned} \quad (12)$$

Each update costs a handful of dense  $N \times N$  matrix products plus one  $LU$ -based solve.

## 3 Structure of homogeneous layers

### 3.1 A per-harmonic $2 \times 2$ block lemma

**Lemma 1.** *Let the  $2n$  field components be ordered by harmonic as  $[E_{x,1}, E_{y,1}, E_{x,2}, E_{y,2}, \dots, E_{x,n}, E_{y,n}]$  (the per-harmonic interleaving). Then for a homogeneous isotropic layer of permittivity  $\varepsilon_0$ ,*

$$\mathbf{K} = \text{blkdiag}(\mathbf{K}^{(1)}, \dots, \mathbf{K}^{(n)}), \quad \mathbf{K}^{(h)} = \omega^2 \mathbf{I}_2 - \frac{1}{\varepsilon_0} \begin{bmatrix} k_{y,h}^2 & -k_{x,h} k_{y,h} \\ -k_{x,h} k_{y,h} & k_{x,h}^2 \end{bmatrix}, \quad (13)$$

the propagation constants satisfy  $\mathbf{q}_h = \mathbf{q}_{n+h} = q_h$  with  $q_h^2 = \varepsilon_0 \omega^2 - k_{x,h}^2 - k_{y,h}^2$ , and  $\Phi = \mathbf{I}_{2n}$ .

*Proof.* The proof is a direct computation. From (10) with  $\mathbf{E} = 1/\varepsilon_0$  and  $\mathbf{J}_k = [-\text{diag}(\mathbf{k}_y); \text{diag}(\mathbf{k}_x)]$ , the product

$$\mathbf{J}_k \mathbf{J}_k^\top = \begin{bmatrix} -\text{diag}(\mathbf{k}_y) \\ \text{diag}(\mathbf{k}_x) \end{bmatrix} \begin{bmatrix} -\text{diag}(\mathbf{k}_y) & \text{diag}(\mathbf{k}_x) \end{bmatrix} = \begin{bmatrix} \text{diag}(\mathbf{k}_y^2) & -\text{diag}(\mathbf{k}_x \mathbf{k}_y) \\ -\text{diag}(\mathbf{k}_x \mathbf{k}_y) & \text{diag}(\mathbf{k}_x^2) \end{bmatrix} \quad (14)$$

is a  $2 \times 2$  block matrix whose blocks are themselves *diagonal*. In the standard  $[E_{x,1} \dots E_{x,n}; E_{y,1} \dots E_{y,n}]$  ordering, each harmonic  $h$  therefore couples only to itself: the  $(h, h)$  entry of the  $(1, 1)$ -block, the  $(h, n+h)$  entry of the  $(1, 2)$ -block, etc. Forming the  $2 \times 2$  submatrix of rows/columns  $\{h, n+h\}$  for each  $h$  and applying the interleaving permutation that orders the components as  $[E_{x,1}, E_{y,1}, E_{x,2}, E_{y,2}, \dots]$  yields exactly (13); different harmonics are never coupled.

It remains to verify the eigensystem. Because (13) is block diagonal, each harmonic obeys an independent dispersion relation. Writing the modal fields as  $e^{i(q_h z)} e^{i(k_{x,h} x + k_{y,h} y)}$  and inserting into Maxwell's equations (2) with  $\varepsilon = \varepsilon_0$  gives  $k_{x,h}^2 + k_{y,h}^2 + q_h^2 = \varepsilon_0 \omega^2$ , i.e.  $q_h^2 = \varepsilon_0 \omega^2 - k_{x,h}^2 - k_{y,h}^2$ . The corresponding eigenvectors are the

coordinate vectors, so in the interleaved ordering  $\Phi = \mathbf{I}_{2n}$  and the propagation constants  $\mathbf{q} = [\mathbf{q}_x; \mathbf{q}_y]$  satisfy  $\mathbf{q}_x = \mathbf{q}_y = q_h$  componentwise.  $\square$

The lemma is exact for *any* harmonic set — rectangular, circular, or anisotropic  $(n_x, n_y)$  — and for *any* incidence, including oblique ( $k_{y0} \neq 0$ ): the only inputs are the per-harmonic pairs  $(k_{x,h}, k_{y,h})$ . It is the reason the quasi-1D “diagonal core” tricks known in the literature for  $k_y = 0$  generalize directly to full 2D.

**Corollary 1.** *The interface matrices (11) between two homogeneous layers, and therefore the scattering matrix of any all-homogeneous stack, are  $2 \times 2$ -block-diagonal in the interleaved ordering.*

*Proof.* For two homogeneous layers  $\Phi_\ell = \Phi_{\ell+1} = \mathbf{I}_{2n}$ , so the first interface matrix in (11) is  $\mathbf{Q}_\ell = \mathbf{I}_{2n}$ . The second is  $\mathbf{P}_\ell = \text{diag}(\mathbf{q}_\ell) \mathbf{K}_\ell^{-1} \mathbf{K}_{\ell+1} \text{diag}(1/\mathbf{q}_{\ell+1})$ . By Lemma 1 each of the four factors is block-diagonal with  $2 \times 2$  blocks (the diag factors are diagonal, hence block-diagonal), and products and inverses of block-diagonal matrices are block-diagonal. Per harmonic  $h$ ,

$$\mathbf{P}^{(h)} = \frac{q_{\ell,h}}{q_{\ell+1,h}} (\mathbf{K}_\ell^{(h)})^{-1} \mathbf{K}_{\ell+1}^{(h)}, \quad (15)$$

with  $\mathbf{K}^{(h)}$  as in (13). The transmission matrices  $\mathbf{T}_{11} = \frac{1}{2}(\mathbf{I} + \mathbf{P})$ ,  $\mathbf{T}_{12} = \frac{1}{2}(\mathbf{I} - \mathbf{P})$ , the diagonal phase factors  $\mathbf{d}_\ell$ , and every matrix in the star update (12) are then block-diagonal as well: the inverse  $(\mathbf{T}_{11} - \mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{12})^{-1}$  of a block-diagonal matrix is block-diagonal, and all products preserve the structure. By induction over the recursion (12), the  $S$ -matrix of any all-homogeneous range is  $2 \times 2$ -block-diagonal.  $\square$

Each harmonic’s  $2 \times 2$  recursion is thus fully independent of the others; only the per-harmonic pairs  $(k_{x,h}, k_{y,h})$  enter, through (13).

### 3.2 The per-harmonic uniform-suffix algorithm

Consider a stack split at the start  $m$  of its trailing all-uniform suffix  $[m, L)$ , i.e. layers  $m, \dots, L-1$  are homogeneous. Corollary 1 suggests the following exact factorization.

1. *Prefix.* Compute  $\mathbf{L} = S(0, m)$  by the general dense star recursion (12) over the (few) layers that contain the pattern, including the boundary homogeneous layer  $m-1$ .
2. *Suffix.* Compute the  $S$ -matrix of the all-uniform range  $[m, L)$  by the per-harmonic  $2 \times 2$  recursion

$$\begin{aligned} \mathbf{P}^{(h)} &= \frac{q_{\ell,h}}{q_{\ell+1,h}} (\mathbf{K}_\ell^{(h)})^{-1} \mathbf{K}_{\ell+1}^{(h)}, \quad \mathbf{T}_{11}^{(h)} = \frac{1}{2}(\mathbf{I}_2 + \mathbf{P}^{(h)}), \quad \mathbf{T}_{12}^{(h)} = \frac{1}{2}(\mathbf{I}_2 - \mathbf{P}^{(h)}), \\ \mathbf{M}^{(h)} &= (\mathbf{T}_{11}^{(h)} - d_{1,h} \mathbf{S}_{12}^{(h)} \mathbf{T}_{12}^{(h)})^{-1}, \quad \mathbf{S}_{11}^{(h)} = d_{1,h} \mathbf{S}_{11}^{(h)} \mathbf{M}^{(h)}, \dots \end{aligned} \quad (16)$$

for each harmonic  $h = 1, \dots, n$ , storing the four  $2 \times 2$  blocks per harmonic. The suffix is thereby built in  $O(n(L-m))$  operations, each involving a  $2 \times 2$  inverse.

3. *Assembly.* Join the dense prefix and the block-diagonal suffix by the overlapping cascade (17) below ( $\mathbf{L}$  and  $\mathbf{R}$  share the boundary layer  $m$ ).

$$\mathbf{M} = (\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}, \quad \mathbf{S}_{11} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{11}, \quad \mathbf{S}_{12} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{R}_{12}, \quad \dots \quad (17)$$

**Derivation of the overlapping cascade.** With the scattering convention  $[a_{\text{out}}; b_{\text{in}}] = S[a_{\text{in}}; b_{\text{out}}]$ , the prefix  $\mathbf{L} = S(0, m)$  and the suffix  $\mathbf{R} = S(m, L)$  both reference the amplitudes  $(a_m, b_m)$  on the shared layer  $m$ :

$$a_m = \mathbf{L}_{11} a_0 + \mathbf{L}_{12} b_m, \quad b_m = \mathbf{R}_{21} a_m + \mathbf{R}_{22} b_L, \quad a_L = \mathbf{R}_{11} a_m + \mathbf{R}_{12} b_L. \quad (18)$$

Eliminating  $a_m, b_m$  between the first two relations gives  $(\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21}) a_m = \mathbf{L}_{11} a_0 + \mathbf{L}_{12} \mathbf{R}_{22} b_L$ , i.e.  $a_m = \mathbf{M}(\mathbf{L}_{11} a_0 + \mathbf{L}_{12} \mathbf{R}_{22} b_L)$  with  $\mathbf{M} = (\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}$ . Substituting into  $a_L$  yields  $\mathbf{S}_{11} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{11}$  and  $\mathbf{S}_{12} =$

Table 1: Cost of the  $S$ -matrix of a stack with a patterned prefix and an  $N_{\text{layers}}$ -layer uniform suffix ( $N = 2n$ ).

|                    | generic                       | per-harmonic suffix      |
|--------------------|-------------------------------|--------------------------|
| uniform interfaces | $O((2n)^3 N_{\text{layers}})$ | $O(n N_{\text{layers}})$ |
| patterned prefix   | $O((2n)^3)$ per layer         | same                     |
| assembly           | $O((2n)^3)$                   | $O((2n)^3)$ (once)       |

$\mathbf{R}_{11}\mathbf{M}\mathbf{L}_{12}\mathbf{R}_{22} + \mathbf{R}_{12}$ ; the backward equations give  $\mathbf{S}_{21} = \mathbf{L}_{21} + \mathbf{L}_{22}\mathbf{R}_{21}\mathbf{M}\mathbf{L}_{11}$  and  $\mathbf{S}_{22} = \mathbf{L}_{22}\mathbf{R}_{21}\mathbf{M}\mathbf{L}_{12}\mathbf{R}_{22} + \mathbf{L}_{22}\mathbf{R}_{22}$ . Because  $\mathbf{R}_{21}, \mathbf{R}_{22}$  are  $2 \times 2$ -block-diagonal, the products  $\mathbf{L}_{12}\mathbf{R}_{21}$  and  $\mathbf{R}_{11}(\cdot)$  are  $O(n)$  block scalings rather than dense products; the only dense  $O((2n)^3)$  operations in the whole suffix assembly are the single matrix inverse  $(\mathbf{I} - \mathbf{L}_{12}\mathbf{R}_{21})^{-1}$  and the two dense products  $\mathbf{M}\mathbf{L}_{11}$ ,  $\mathbf{M}(\mathbf{L}_{12}\mathbf{R}_{22})$ .

Table 1 summarizes the scaling. For a stack whose patterned layers are few (as in the EUV-mask geometry: vacuum | absorber | cap |  $40 \times (\text{Mo/Si})$  | substrate), the entire multilayer suffix is reduced from 80 dense  $O((2n)^3)$  star products to  $O(n \cdot 82)$  scalar- $2 \times 2$  operations, and the residual dense work is one prefix and one cascade.

### 3.3 Relation to quasi-1D and periodic-stack methods

When the harmonic set is the  $x$ -only row and  $k_{y,h} = 0$  for all  $h$ , the blocks (13) are diagonal and (16) reduces to independent *scalar* Fresnel-like recursions per harmonic — the classical quasi-1D trick. When the suffix is periodic (a repeated bilayer), the per-harmonic recursion handles it as cheaply as the periodic cores exponentiated by the Redheffer cascade; no periodicity assumption is required. The factorization therefore subsumes both the quasi-1D diagonal core and periodic binary exponentiation as special cases.

## 4 Exact decompositions of the patterned-layer eigenproblem

The remaining  $O((2n)^3)$  bottleneck is the eigenproblem (8) of the patterned layer. We show that for the two most common structures it decouples exactly into two  $n \times n$  problems, at  $4 \times$  fewer eigensolver flops, with a constructive formula for the eigenvectors. Throughout, the harmonic set is the 1D row of a  $y$ -invariant grating, so that  $\mathbf{k}_y = \kappa_y \mathbf{1}$  is a constant vector, but we emphasize that the block-diagonal case also covers certain 2D patterns (see Section 10).

### 4.1 Block-diagonal $\mathbf{M}$ : normal incidence

At normal incidence  $k_{x0} = k_{y0} = 0$  and for a  $y$ -invariant profile  $\mathbf{k}_y = \mathbf{0}$ , the block  $\mathbf{J}_k = [-\text{diag}(\mathbf{k}_y); \text{diag}(\mathbf{k}_x)]$  reduces to  $\mathbf{J}_k = \begin{bmatrix} \mathbf{0} \\ \text{diag}(\mathbf{k}_x) \end{bmatrix}$ , and (10) gives, writing  $D_x = \text{diag}(\mathbf{k}_x)$ ,

$$\mathbf{J}_k \mathbf{E} = \begin{bmatrix} \mathbf{0} \\ D_x \mathbf{E} \end{bmatrix}, \quad (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & D_x \mathbf{E} D_x \end{bmatrix}, \quad (19)$$

so that  $\mathbf{K} = \text{diag}(\omega^2 \mathbf{I}, \omega^2 \mathbf{I} - D_x \mathbf{E} D_x)$ . Similarly  $\mathbf{k} = [D_x; \mathbf{0}]$  gives  $\mathbf{k} \mathbf{k}^\top = \text{diag}(D_x^2, \mathbf{0})$ , and  $\text{ep2} = \text{blkdiag}(\mathcal{C}, \mathcal{C})$ . Hence

$$\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top = \begin{bmatrix} \mathcal{C} \omega^2 - D_x^2 & \mathbf{0} \\ \mathbf{0} & \mathcal{C}(\omega^2 \mathbf{I} - D_x \mathbf{E} D_x) \end{bmatrix} \equiv \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{22} \end{bmatrix}, \quad (20)$$

with, using  $\mathcal{C} \mathbf{E} = \mathbf{I}$ ,

$$\mathbf{M}_{11} = \omega^2 \mathcal{C} - D_x^2, \quad \mathbf{M}_{22} = \omega^2 \mathcal{C} - \mathcal{C} D_x \mathcal{C}^{-1} D_x. \quad (21)$$

The  $2n \times 2n$  eigenproblem (8) therefore decouples exactly into two  $n \times n$  problems,  $\text{eig}(\mathbf{M}) = \text{eig}(\mathbf{M}_{11}) \cup \text{eig}(\mathbf{M}_{22})$ ,  $\Phi = \text{blkdiag}(\Phi_1, \Phi_2)$ , reducing the eigensolver flops by a factor four. (For a D4-symmetric 2D pattern at normal incidence the off-diagonal blocks also vanish exactly, and the same decomposition applies.)

## 4.2 Block-lower-triangular $\mathbf{M}$ : oblique incidence

At oblique incidence  $k_{y0} \neq 0$  the 1D harmonic set still has  $\mathbf{k}_y = \kappa_y \mathbf{1}$  (a constant vector), so that  $D_y = \text{diag}(\mathbf{k}_y) = \kappa_y \mathbf{I}$  and  $\mathbf{J}_k = [-\kappa_y \mathbf{I}; D_x]$ . Evaluating (10)–(9) blockwise,

$$\mathbf{J}_k \mathbf{E} = \begin{bmatrix} -\kappa_y \mathbf{E} \\ D_x \mathbf{E} \end{bmatrix}, \quad (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top = \begin{bmatrix} \kappa_y^2 \mathbf{E} & -\kappa_y \mathbf{E} D_x \\ -\kappa_y D_x \mathbf{E} & D_x \mathbf{E} D_x \end{bmatrix}, \quad (22)$$

so that

$$\mathbf{K} = \begin{bmatrix} \omega^2 \mathbf{I} - \kappa_y^2 \mathbf{E} & \kappa_y \mathbf{E} D_x \\ \kappa_y D_x \mathbf{E} & \omega^2 \mathbf{I} - D_x \mathbf{E} D_x \end{bmatrix}, \quad \mathbf{k} \mathbf{k}^\top = \begin{bmatrix} D_x^2 & \kappa_y D_x \\ \kappa_y D_x & \kappa_y^2 \mathbf{I} \end{bmatrix}. \quad (23)$$

The four blocks of  $\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top$  are

$$\mathbf{M}_{11} = \mathcal{C}(\omega^2 \mathbf{I} - \kappa_y^2 \mathbf{E}) - D_x^2 = \omega^2 \mathcal{C} - \kappa_y^2 \mathbf{I} - D_x^2, \quad (24)$$

$$\mathbf{M}_{12} = \mathcal{C} \kappa_y \mathbf{E} D_x - \kappa_y D_x = \kappa_y (\mathcal{C} \mathbf{E} - \mathbf{I}) D_x = \mathbf{0}, \quad (25)$$

$$\mathbf{M}_{21} = \mathcal{C} \kappa_y D_x \mathbf{E} - \kappa_y D_x = \kappa_y (\mathcal{C} D_x \mathcal{C}^{-1} - D_x) \neq \mathbf{0}, \quad (26)$$

$$\mathbf{M}_{22} = \mathcal{C}(\omega^2 \mathbf{I} - D_x \mathbf{E} D_x) - \kappa_y^2 \mathbf{I} = \omega^2 \mathcal{C} - \kappa_y^2 \mathbf{I} - \mathcal{C} D_x \mathcal{C}^{-1} D_x. \quad (27)$$

Because  $\mathbf{M}_{12} = \mathbf{0}$ ,  $\mathbf{M}$  is block-lower-triangular,

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix}, \quad (28)$$

and  $\det(\mathbf{M} - \lambda \mathbf{I}) = \det(\mathbf{M}_{11} - \lambda \mathbf{I}) \det(\mathbf{M}_{22} - \lambda \mathbf{I})$ : the eigenvalues split exactly into two  $n$ -dimensional problems  $\text{eig}(\mathbf{M}) = \text{eig}(\mathbf{M}_{11}) \cup \text{eig}(\mathbf{M}_{22})$ , a  $4\times$  reduction in eigensolver flops. The eigenvectors, however, remain coupled through the nonzero lower-left block  $\mathbf{M}_{21}$ ; their exact reconstruction is the subject of the next subsection.

## 4.3 Exact reconstruction of the coupled eigenvectors

Because  $\mathbf{M}$  is block-lower-triangular, the eigenvectors of the two eigenvalue families are constructed as follows.

*Columns for  $\lambda_{A,j} = \text{eig}_j(\mathbf{M}_{11})$ .* Let  $\mathbf{M}_{11} = \Phi_A \Lambda_A \Phi_A^{-1}$  with columns  $\Phi_A = [\mathbf{a}_1, \dots, \mathbf{a}_n]$  and  $\mathbf{M}_{11} \mathbf{a}_j = \lambda_{A,j} \mathbf{a}_j$ , and let  $\mathbf{M}_{22} = \Phi_C \Lambda_C \Phi_C^{-1}$ . Writing an eigenvector as  $[\mathbf{a}_j; \mathbf{v}_j]$ , the second block row of  $(\mathbf{M} - \lambda_{A,j} \mathbf{I})[\mathbf{a}_j; \mathbf{v}_j] = \mathbf{0}$  reads

$$(\mathbf{M}_{22} - \lambda_{A,j} \mathbf{I}) \mathbf{v}_j = -\mathbf{M}_{21} \mathbf{a}_j, \quad (29)$$

the first block row being satisfied identically. When  $\lambda_{A,j} \notin \text{eig}(\mathbf{M}_{22})$  the resolvent of  $\mathbf{M}_{22}$  gives

$$\mathbf{v}_j = -(\mathbf{M}_{22} - \lambda_{A,j} \mathbf{I})^{-1} \mathbf{M}_{21} \mathbf{a}_j = -\Phi_C \text{diag}\left(\frac{1}{\lambda_{C,i} - \lambda_{A,j}}\right) \Phi_C^{-1} \mathbf{M}_{21} \mathbf{a}_j. \quad (30)$$

*Columns for  $\lambda_{C,j} = \text{eig}_j(\mathbf{M}_{22})$ .* Writing an eigenvector as  $[\mathbf{u}_j; \mathbf{c}_j]$ , the second block row  $(\mathbf{M}_{22} - \lambda_{C,j} \mathbf{I}) \mathbf{c}_j = \mathbf{0}$  is satisfied by  $\mathbf{c}_j = \Phi_C[:, j]$ ; the first block row  $(\mathbf{M}_{11} - \lambda_{C,j} \mathbf{I}) \mathbf{u}_j = \mathbf{0}$  then forces  $\mathbf{u}_j = \mathbf{0}$  provided  $\lambda_{C,j} \notin \text{eig}(\mathbf{M}_{11})$ . The corresponding modal columns are therefore  $[\mathbf{0}; \Phi_C[:, j]]$ .

Collecting both families gives the block-lower-triangular modal matrix

$$\Phi = \begin{bmatrix} \Phi_A & \mathbf{0} \\ \mathbf{V} & \Phi_C \end{bmatrix}, \quad \mathbf{V} = -\Phi_C \left( (\Phi_C^{-1} \mathbf{M}_{21} \Phi_A) \oslash \mathbf{D} \right), \quad \mathbf{D}_{ij} = \lambda_{C,i} - \lambda_{A,j}, \quad (31)$$

where  $\oslash$  denotes element-wise (Hadamard) division; the  $i$ -th row of  $(\Phi_C^{-1} \mathbf{M}_{21} \Phi_A) \oslash \mathbf{D}$  is  $(\Phi_C^{-1} \mathbf{M}_{21} \Phi_A)_{ij} / D_{ij}$ , which is exactly  $(\Phi_C^{-1} \mathbf{M}_{21} \mathbf{a}_j)_i / (\lambda_{C,i} - \lambda_{A,j})$ , matching (30) column by column. The reconstruction (31) costs two  $n \times n$  eigendecompositions, one  $n \times n$  solve  $(\Phi_C^{-1} \mathbf{M}_{21} \Phi_A)$ , and two  $O(n^3)$  products, versus one  $2n \times 2n$  eigendecomposition; the results are numerically identical to the full solve (verified to 17 digits in the scattering

observables). If a pair of eigenvalues becomes exactly degenerate ( $\min_{ij} |\mathbf{D}_{ij}| \rightarrow 0$ ) the resolvent (30) is singular and the code falls back to the full  $2n \times 2n$  solve; the detection is exact (off-diagonal-block norm test), so the result is never silently wrong.

## 5 TE/TM-decoupled $S$ -matrix path

At normal incidence with block-diagonal  $\mathbf{K}, \Phi, \mathbf{q}$  (Section 4.1), every interface matrix (11) is block-diagonal with two  $n \times n$  blocks (the TE and TM polarizations). The Redheffer recursion and the cascade then run independently per polarization on  $n \times n$  blocks — a further  $4\times$  reduction on the (already reduced) patterned prefix and the assembly — and the result is reassembled as  $\mathbf{S} = \text{blkdiag}(\mathbf{S}_{\text{TE}}, \mathbf{S}_{\text{TM}})$ . Combined with the per-harmonic suffix of Section 3.2, the quasi-1D normal-incidence solve is thus scalar per harmonic throughout the uniform suffix and  $n \times n$ -block in the prefix.

## 6 Binary gratings: low-rank permittivity structure

For the step-rect profile  $\varepsilon(x) = \varepsilon_1$  on  $|x| < x_1$ ,  $\varepsilon_2$  otherwise, the Fourier coefficients follow from a one-line integral. Writing  $\varepsilon(x) = \varepsilon_2 + \Delta\varepsilon \chi(x)$  with  $\Delta\varepsilon = \varepsilon_1 - \varepsilon_2$  and  $\chi$  the indicator of  $|x| < x_1$ ,

$$\hat{\varepsilon}_m = \frac{1}{L_x} \int_{-L_x/2}^{L_x/2} \varepsilon(x) e^{-i2\pi mx/L_x} dx = \begin{cases} \varepsilon_2 + \Delta\varepsilon f, & m = 0, \\ \Delta\varepsilon f \text{sinc}(mf), & m \neq 0, \end{cases} \quad (32)$$

where  $f = 2x_1/L_x$  is the fill fraction,  $\text{sinc}(z) = \sin(\pi z)/(\pi z)$ , and the second line uses  $\int_{-x_1}^{x_1} e^{-i\beta x} dx = 2\sin(\beta x_1)/\beta$  with  $\beta = 2\pi m/L_x$ . The Toeplitz convolution matrix is therefore  $\mathcal{C} = \varepsilon_2 \mathbf{I} + \Delta\varepsilon f \mathbf{S}$  with the “discrete sinc” matrix  $\mathbf{S}_{ij} = \text{sinc}(f(i-j))$ .

The matrix  $\mathbf{S}$  is the finite-dimensional prolate (Slepian) matrix [Sle78]: its symbol is the indicator of the frequency band  $|\theta| \leq \pi f$ , and its eigenvalues obey the two-interval concentration law — approximately  $nf$  eigenvalues near 1 and the rest exponentially small,

$$\text{rank}_\tau(\mathbf{S}) \approx nf + O(\log \tau^{-1}). \quad (33)$$

For the EUV example of Section 9 ( $f = 500/3500 \approx 0.143$ ,  $n = 253$ ) this gives a numerical rank of  $\approx 52$  at  $\tau = 10^{-14}$ , i.e.  $\mathbf{S}$  is  $5\times$  rank-deficient; a rank-52 reconstruction reproduces  $\mathbf{S}$  to  $\sim 1.5 \times 10^{-15}$ . Two consequences follow.

First, *the analytic coefficients* (32) replace the grid sampling and fast Fourier transform in the construction of  $\mathcal{C}$ , and remove the pixelation error of a discrete grid (a 1000-pixel cell quantizes the fill to  $f = 0.1420$  against the exact  $f = 0.142857$ ). For smooth profiles (Gaussian, Lorentzian) the coefficients decay exponentially and the harmonic count required for a fixed accuracy drops correspondingly.

Second, *both patterned-layer eigenproblems become diagonal-plus-low-rank*. Let  $\mathbf{S} = U\Sigma U^\top$  be the (numerically) rank- $r$  eigendecomposition,  $U \in \mathbb{C}^{n \times r}$ ,  $\Sigma \in \mathbb{R}^{r \times r}$ . By the Woodbury identity

$$\mathcal{C}^{-1} = (\varepsilon_2 \mathbf{I} + \Delta\varepsilon f U \Sigma U^\top)^{-1} = \frac{1}{\varepsilon_2} \left( \mathbf{I} - U (\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \Sigma)^{-1} (\Delta\varepsilon f \Sigma) U^\top \right), \quad (34)$$

i.e. a diagonal plus a rank- $r$  update, with the abbreviation  $\Gamma = (\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \Sigma)^{-1} \Delta\varepsilon f \Sigma$ . Using  $D = \text{diag}(\mathbf{k}_x)$ ,

$$\mathcal{C} D \mathcal{C}^{-1} D = D^2 + \frac{1}{\varepsilon_2} \left( \Delta\varepsilon f \mathbf{S} D^2 - \varepsilon_2 D U T U^\top D - \Delta\varepsilon f \mathbf{S} D U T U^\top D \right), \quad (35)$$

each bracketed term having rank at most  $r$  (a product of the rank- $r$  factor  $U T U^\top$  or  $U \Sigma U^\top$  with diagonal

matrices). Hence the patterned blocks (21) take the form

$$\mathbf{M}_{11} = \text{diag}(\beta_h) + \omega^2 \Delta \varepsilon f \mathbf{S}, \quad \mathbf{M}_{22} = \text{diag}(\beta'_h) + \text{rank-} \leq 4r \text{ update}, \quad (36)$$

with  $\beta_h = \omega^2 \varepsilon_2 - \kappa_y^2 - k_{x,h}^2$ . For lossless media both matrices are real symmetric, and a diagonal-plus-low-rank (DPLR) eigensolver [GV13] computes all eigenvalues in  $O(nr + r^3)$  instead of  $O(n^3)$ : the active subspace is only  $r$ -dimensional, and the bulk eigenvalues are small corrections to the diagonal entries. For lossy media the matrices are complex symmetric and the secular-equation approach applies with a complex root finder; we report the lossless speedup below and leave the complex case as a straightforward extension.

## 7 Block-wise post-processing

The absorbed power in a layer is  $\text{val} = \text{tr}(\mathbf{abM}\mathbf{F}^\dagger \mathbf{M}_{\text{tot}} \mathbf{F})$ , where  $\mathbf{ab} = [\mathbf{a}_i; \mathbf{b}_i] \in \mathbb{C}^{4n}$  stacks the forward and backward modal amplitudes,  $\mathbf{abM} = \mathbf{abab}^\dagger \odot \mathbf{M}_t$  is a Hadamard (element-wise) product with the modal  $z$ -integral matrix  $\mathbf{M}_t$ , and  $\mathbf{M}_{\text{tot}}$  is the block-diagonal material weighting. A direct evaluation materializes the  $4n \times 4n$  matrices  $\mathbf{M}_t, \mathbf{abM}$  and the  $3n \times 3n$  matrices  $\mathbf{M}_{\text{tot}}, \mathbf{F}$ .

Using the block forms

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_{xy} & -\mathbf{F}_{xy} \\ \mathbf{F}_{az} & \mathbf{F}_{az} \end{bmatrix}, \quad \mathbf{M}_t = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{M}_{ab} \\ \mathbf{M}_{ab} & \mathbf{M}_{aa} \end{bmatrix}, \quad (37)$$

the interior product collapses onto  $2n \times 2n$  blocks. Writing  $\mathbf{T} = \mathbf{F}^\dagger \mathbf{M}_{\text{tot}} \mathbf{F}$  with  $\mathbf{M}_{\text{tot}} = \text{blkdiag}(\mathbf{M}_{xy}, \mathbf{M}_z)$ ,

$$\mathbf{T} = \begin{bmatrix} \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} & -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} \\ -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} & \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{bmatrix}, \quad (38)$$

with  $\mathbf{A} = \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az}$  and  $\mathbf{B} = -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az}$ . The trace is then expanded without ever forming  $\mathbf{abM}$ :

$$\text{val} = \text{tr}((\mathbf{abab}^\dagger \odot \mathbf{M}_t) \mathbf{T}) = \sum_{i,j} \text{ab}_i (\mathbf{M}_t \odot \mathbf{T}^\top)_{ij} \overline{\text{ab}_j} = \text{ab}^\top (\mathbf{M}_t \odot \mathbf{T}^\top) \overline{\text{ab}}, \quad (39)$$

using  $(\mathbf{X} \odot \mathbf{Y})_{ij} = \mathbf{X}_{ij} \mathbf{Y}_{ij}$  and  $\text{tr}(\mathbf{Z}) = \sum_i \mathbf{Z}_{ii}$ . Blockwise, with  $\mathbf{W}_A = \mathbf{M}_{aa} \circ \mathbf{A}^\top$  and  $\mathbf{W}_B = \mathbf{M}_{ab} \circ \mathbf{B}^\top$ ,

$$\text{val} = \mathbf{a}^\top \mathbf{W}_A \bar{\mathbf{a}} + \mathbf{b}^\top \mathbf{W}_A \bar{\mathbf{b}} + \mathbf{a}^\top \mathbf{W}_B \bar{\mathbf{b}} + \mathbf{b}^\top \mathbf{W}_B \bar{\mathbf{a}}. \quad (40)$$

The evaluation therefore needs two  $2n \times 2n$  Hadamard products  $(\mathbf{W}_A, \mathbf{W}_B)$ , the four block products that form  $\mathbf{A}, \mathbf{B}$  in (38), and four matvecs — never materializing the  $4n \times 4n$  and  $3n \times 3n$  matrices. This is a  $\sim 20\times$  reduction in transient memory for the absorption and stress-tensor integrals, at no change in the result (identical to the original formula to  $\sim 10^{-15}$ ).

## 8 Implementation

The reductions above are implemented in a C++17 library using Eigen, FFTW, and LAPACK/BLAS. The remaining algorithmic considerations are:

- *Parallel eigenpairs.* The two independent  $n \times n$  eigenproblems of Sections 4.1–4.2 are solved concurrently on two threads. OpenBLAS executes `zgeev` single-threaded (LAPACK-serial), so the pair runs  $\sim 1.5$ – $1.8\times$  faster with bit-identical results; with a threaded LAPACK (e.g. MKL) the parallelization is disabled by a flag.
- *Star-product details.* The phase factors  $\mathbf{d}_1, \mathbf{d}_2$  are kept as vectors and applied as diagonal scalings ( $O(n)$  per product); matrix inversions in the star update are replaced by one *LU* factorization plus two back-

Table 2: Validation against grcwa / S4.

| case                   | $R, T$ (this work) | reference              | match       |
|------------------------|--------------------|------------------------|-------------|
| S4 golden $T$ , p-pol  | 0.853173           | 0.852499*              | $< 10^{-3}$ |
| S4 golden $T$ , s-pol  | 0.839654           | 0.839005*              | $< 10^{-3}$ |
| ex1 square holes       | 0.132415, 0.867585 | 0.13241493, 0.86758507 | ✓           |
| ex2 two layers         | 0.132599, 0.867401 | 0.13259894, 0.86740106 | ✓           |
| ex4 hexagonal          | 0.164450, 0.835550 | 0.16445004, 0.83554996 | ✓           |
| EUV mirror 13.5 nm $R$ | 0.599171           | 0.599171               | exact       |
| EUV absorber $R$       | 0.578759           | 0.578790               | exact       |

\* S4 golden value embedded in the test; grcwa itself computes 0.853173 / 0.839654, reproduced exactly by this work.

Table 3: Full-2D EUV absorber,  $S$ -matrix (‘rt\_solve’) before/after the per-harmonic uniform suffix (machine under load;  $R$  identical in both columns).

| $n$ | generic | per-harmonic suffix | speedup     |
|-----|---------|---------------------|-------------|
| 201 | 1354 ms | 369 ms              | $3.7\times$ |
| 301 | 3268 ms | 964 ms              | $3.4\times$ |

substitutions; the products  $\mathbf{d}_1\mathbf{S}_{12}$  and  $\mathbf{S}_{22}\mathbf{T}_{12}$  are hoisted as common subexpressions; and the LAPACK workspace is reused thread-locally.

- *Caching.* FFTW plans are cached per grid size; uniform layers with identical permittivity share their  $(\mathbf{K}, \mathbf{q}, \Phi)$ ; identical uniform–uniform interface matrices are reused; and the most recent  $S$ -matrix is memoized.
- *Memory.* Uniform layers deduplicate their operators, and the block-wise post-processing of Section 7 keeps the peak footprint bounded.

## 9 Results

### 9.1 Validation

Table 2 compares the solver against the reference implementations grcwa [] and S4 [LF14]. The S4 golden test [LF14] (two-layer hole pattern) is reproduced within the  $10^{-3}$  tolerance of the reference, and the grcwa examples are matched to printed precision; the EUV cases agree with grcwa to all printed digits. Per-order reflected-field coefficients and real-space fields agree with grcwa to  $\sim 10^{-13}$ .

The EUV absorber convergence with the harmonic count is  $R = 0.5668, 0.5788, 0.5791, 0.57925$  for  $n = 9, 97, 197, 301$  respectively — the per-harmonic suffix keeps the solve cost  $O(n)$  in the number of uniform layers at every truncation.

### 9.2 Performance

Table 3 reports the effect of the per-harmonic uniform suffix (Section 3.2) on the full-2D EUV absorber (vacuum | TaN absorber | Ru cap |  $40\times(\text{Mo/Si})$  | Si substrate). The suffix alone accelerates the  $S$ -matrix assembly by  $3.4\text{--}3.7\times$  with bit-identical reflectivity; the gain grows with  $n$  because the suffix cost is now linear in  $n$  while the reference dense products scale as  $(2n)^3$ .

On the quasi-1D geometry (the same absorber as a  $y$ -invariant bar,  $L_x = 3.5\,\mu\text{m}$ ,  $n = 113$ ) the combined reductions — 1D harmonic filter, per-harmonic suffix, and TE/TM decoupling — give an end-to-end speedup of  $\sim 30\times$  over the reference grcwa at identical reflectivity  $R = 0.528927$ . The parallel eigenpairs additionally reduce the patterned-layer setup by  $\sim 1.5\text{--}1.8\times$ .

## 10 Discussion

The per-harmonic uniform-suffix factorization and the exact eigenproblem decompositions are exact: they change only the *order* of arithmetic, not the model, and the scattering observables are bit-identical to the generic implementation. Three caveats bound their applicability.

First, the block-lower-triangular decomposition (Section 4.2) requires the harmonic set to be the 1D row of a  $y$ -invariant grating (constant  $\mathbf{k}_y$ ). For genuinely 2D patterns at oblique incidence,  $\mathbf{M}_{12} = \mathcal{C}\text{diag}(\mathbf{k}_y)\text{Ediag}(\mathbf{k}_x) - \text{diag}(\mathbf{k}_x\mathbf{k}_y) \neq \mathbf{0}$ , and the code falls back to the full  $2n \times 2n$  solve; the detection is numerical (norm test on the off-diagonal blocks) so no wrong results are possible.

Second, the block-diagonal decomposition (Section 4.1) applies at normal incidence to mirror-symmetric patterns. For asymmetric 2D patterns at oblique incidence neither eig shortcut applies, and the full  $2n \times 2n$  `zgeev` remains — the recognized residual bottleneck. A matrix-free Krylov formulation (preconditioned by the quasi-1D solves above) is the natural next step for very large  $n$ , but is beyond the scope of this paper.

Third, the low-rank structure of Section 6 is fully exploited only for lossless profiles (real symmetric DPLR). For lossy materials the complex-symmetric secular equation requires a complex root finder; this is a straightforward but not yet integrated extension.

Finally, we note that Li’s factorization rules [Li96b] and the normal-vector method [Li97] improve *convergence in  $n$*  for high-contrast metallic gratings (reducing the required  $n$ ), whereas the present reductions reduce the *cost per harmonic*. The two are orthogonal and compound: every harmonic saved by better factorization is processed at the reduced per-harmonic cost.

## 11 Conclusion

We have shown that the dominant cubic cost of rigorous coupled-wave analysis is largely structural. Homogeneous layers decouple per harmonic, making the multilayer stack  $O(n)$ ; the patterned-layer eigenproblem splits exactly at normal and oblique incidence; and binary gratings carry a low-rank permittivity structure. Implemented together, these reductions reproduce the reference solvers to machine precision while accelerating a full-2D EUV mask stack by  $3.5\times$  (multilayer suffix from hours-scale to microseconds) and a quasi-1D stack by  $\sim 30\times$ . The remaining  $O((2n)^3)$  work is confined to the few patterned layers and a single assembly, pointing toward matrix-free Krylov methods as the next scaling step.

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